MedAxis Intelligence — Company Brief & Project Charter

About MedAxis Intelligence

MedAxis Intelligence is a Series C global digital health company headquartered in London, UK, with operations in the United States, Germany, and Singapore. Founded in 2018, MedAxis partners with NHS trusts, private hospital groups, and integrated care systems to deliver AI-powered decision support for clinical operations, patient flow, and care coordination.

MedAxis currently serves 14 hospital networks, 3 integrated care boards, and 2 international private hospital groups. Annual recurring revenue: £38.4M. Team size: 320 employees. Engineering and data science headcount: 74.

The Business Problem

MedAxis has signed a strategic contract with Meridian Health Group — a network of 6 acute hospitals across the UK and Ireland — worth £4.2M over 3 years. The contract requires MedAxis to deliver a production-ready AI decision-support platform within 12 weeks. The platform must:

- Predict 30-day patient readmission risk with clinically acceptable recall (≥ 0.72)
- Forecast hospital bed demand and ICU pressure 14 days ahead
- Predict patient length of stay for ward-level resource planning
- Extract structured information from clinical notes and discharge summaries
- Provide a RAG-powered guideline search assistant for care coordinators
- Be served via a production FastAPI with audit logging and authentication
- Pass a responsible AI review including bias audit, model card, and drift monitoring
- Be presented to Meridian's clinical board with ROI justification

■■ DEADLINE: Week 12 board presentation is contractually fixed. Missing it triggers a £420,000 penalty clause.

Current Metrics — Meridian Health Group

Metric	Current Value	Target	Gap
30-day readmission rate	18.4%	< 14%	–4.4 percentage points
Average length of stay	6.8 days	< 5.5 days	–1.3 days
Bed occupancy (peak)	94.2%	< 88%	–6.2 percentage points
Emergency demand forecast error	±31%	< ±12%	–19 percentage points
Clinical note review time	47 min/patient	< 15 min	–32 min/patient
Delayed discharge rate	22.7%	< 12%	–10.7 percentage points

Financial Stakes

Each percentage point reduction in 30-day readmission rate saves Meridian approximately £310,000 per year in avoided readmission costs. Reducing average length of stay by 1 day across the network frees approximately 1,840 bed-days per year — valued at £2.1M in operational capacity. The total addressable saving from achieving all targets is estimated at £9.2M annually.

■ ***Your models must be good enough to justify clinical deployment. Meridian's Chief Medical Officer has veto power over any model that fails bias review.***

Sprint Timeline

Week	Milestone	Owner
Week 1	Healthcare AI strategy brief, stakeholder register, GitHub repo, success metrics	You
Week 2	PostgreSQL data warehouse, star schema, SQL validation, ADR	You
Week 3	Data quality suite, Great Expectations, feature engineering pipeline	You
Week 4	Readmission model, MLflow tracking, SHAP explanation, model eval report	You
Week 5	Model card, bias audit, threshold memo, clinical safety review	You
Week 6	Demand forecasting, Power BI dashboard, MAPE comparison, capacity memo	You
Week 7	LOS model, resource optimisation, staffing recommendation	You
Week 8	Clinical NLP pipeline, entity extraction, ICD coding prototype	You
Week 9	RAG assistant, vector DB, hallucination testing, Streamlit app	You
Week 10	FastAPI deployment, Docker, test suite, Postman evidence	You
Week 11	MLOps, drift monitoring, GitHub Actions CI/CD, governance pack	You
Week 12	Board deck, ROI model, portfolio pack, STAR interview answers	You

Regulatory Context

MedAxis operates under multiple regulatory frameworks depending on deployment geography. All models deployed in clinical pathways must meet the requirements of the applicable framework. You are responsible for ensuring your deliverables are compliant.

Region	Framework	Key Requirement
UK	MHRA / NHS DTAC	AI as Medical Device classification; DTAC assessment before NHS deployment
UK	UK GDPR / Data Protection Act 2018	Lawful basis for processing health data; DPIA required
EU	EU AI Act (High Risk — Article 6)	Conformity assessment, technical documentation, human oversight
EU	GDPR Article 9	Special category health data — explicit consent or public interest basis
USA	HIPAA	De-identification of PHI; Business Associate Agreement with covered entities
USA	FDA SaMD (Software as Medical Device)	510(k) or De Novo pathway for clinical decision support
Canada	PIPEDA	Consent-based health data processing; breach notification
Singapore	PDPA + MOH AI Guidelines	Data portability; clinical AI governance requirements

■■ **Your model card (Phase 5) must identify which regulatory frameworks apply and how the model meets them. Meridian's legal team will review this before go-live sign-off.**

Glossary

Term	Definition
ADM_TYPE	Admission type: EM = Emergency, EL = Elective, TR = Transfer, MAT = Maternity
DISCH_DEST	Discharge destination: HOME, NH (nursing home), RTH (return to hospital), DIED, OTH
LOS_DAYS	Length of stay in whole days (admission to discharge)
30DR	30-day readmission flag: 1 = readmitted within 30 days of discharge, 0 = not
ICD10_PRI	Primary ICD-10 diagnosis code
ICD10_SEC	Secondary ICD-10 diagnosis code(s), pipe-separated
LAB_FLAG	Abnormal lab result flag: 0 = normal, 1 = mildly abnormal, 2 = critically abnormal
DD_FLAG	Delayed discharge flag: 1 = delayed beyond clinical readiness date
COMORB_CNT	Count of documented comorbidities at admission
PREV_ADM_90	Number of admissions in the 90 days preceding this encounter
DEPRIVATION	IMD decile 1–10 (1 = most deprived, 10 = least deprived)
WARD_CODE	Internal ward identifier — see hospital_reference.pdf for mapping
SITE_ID	Hospital site: MER001–MER006 (Meridian site codes)
ENC_TYPE	Encounter type: IP = inpatient, OP = outpatient, ED = emergency department
SaMD	Software as a Medical Device — regulatory classification for clinical AI
DTAC	Digital Technology Assessment Criteria — NHS procurement standard for digital health
SHAP	SHapley Additive exPlanations — model explainability framework
AUC	Area Under the ROC Curve — model discrimination metric
MAPE	Mean Absolute Percentage Error — forecasting accuracy metric
RAG	Retrieval-Augmented Generation — LLM grounded in a document knowledge base
FHIR	Fast Healthcare Interoperability Resources — HL7 healthcare data standard